

# The Toeplitz Restriction Theorem on Every Second-Countable LCA Group

Automated Math Discovery Candidate Proof

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**Status.** Candidate full solution, likely valid pending expert review. A common  $L^1$  approximate identity replaces the only Euclidean density-point step in the source proof, extending Theorem 7.5 to every second-countable locally compact abelian group.

## Source and Open Problem

Shulman, Todorov, and Turowska [1] study closable Schur multipliers. Section 7 of their paper fixes a second-countable locally compact abelian group  $G$ , with Haar measure  $\mu$ . Their Problem 2 on page 33 asks:

*Does Theorem 7.5 hold for all locally compact abelian groups?*

CLOSABLE MULTIPLIERS

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### 8. OPEN PROBLEMS

In this section we list some open problems. The most important question which we have left unanswered is the following:

**Problem 1.** Is every  $w^*$ -closable multiplier a local Schur multiplier?

**Problem 2.** Does Theorem [7.5](#) hold for all locally compact abelian groups?

**Problem 3.** For which  $f \in C(\mathbb{R})$  is the divided difference  $\check{f}$  a  $w^*$ -closable multiplier?

Since all local multipliers are  $w^*$ -closable, Corollary [6.4](#) shows that a sufficient condition for this to happen is  $\mu(E(f)) = 0$ .

Here is the referenced theorem from page 26 of the source:

For notational simplicity we let  $\Gamma(G) = \Gamma(G, G)$ . We shall frequently use the map  $P : \Gamma(G) \rightarrow A(G)$  given by  $P(f \otimes g)(t) = (\lambda(t)g, \bar{f})$ . Clearly,  $P$  is a surjective contraction.

For a subset  $E \subseteq G$ , let  $E^* = \{(x, y) \in G \times G : x - y \in E\}$ .

**Theorem 7.5.** *Assume that  $G$  is a subgroup of  $\mathbb{R}^n$  or  $\mathbb{T}^n$ ,  $n \in \mathbb{N}$ . Let  $f : G \rightarrow \mathbb{C}$  be a measurable function,  $\varphi = Nf$  and  $U, V \subseteq G$  be measurable sets. The following are equivalent:*

- (i)  $(U \times V) \cap E_f^* \simeq \emptyset$ ;
- (ii)  $\varphi|_{U \times V} \in \mathfrak{S}_{\text{loc}}(U, V)$ ;
- (iii)  $\varphi|_{U \times V} \in \mathfrak{S}_{w^*}(U, V)$ .

We recall the minimum notation needed to state it. For measurable  $f : G \rightarrow \mathbb{C}$ , put

$$Nf(x, y) = f(x - y).$$

The source associates to  $f$  a closed obstruction set  $E_f \subseteq G$ , and writes

$$E_f^* = \{(x, y) \in G \times G : x - y \in E_f\}.$$

A subset of  $U \times V$  is *marginally null* if it is contained in  $(M \times V) \cup (U \times N)$  for null sets  $M \subseteq U$  and  $N \subseteq V$ ; the notation  $\simeq \emptyset$  means marginal nullity. The classes  $\mathfrak{S}_{\text{loc}}(U, V)$  and  $\mathfrak{S}_{w^*}(U, V)$  are, respectively, the local Schur multipliers and the weak-star closable multipliers of the source paper.

For finite-measure measurable  $A, B \subseteq G$ , the elementary tensor  $\mathbf{1}_A \otimes \mathbf{1}_B$  lies in  $\Gamma(G) = L^2(G) \hat{\otimes} L^2(G)$ . The source's contraction  $P : \Gamma(G) \rightarrow A(G)$  satisfies

$$P(\mathbf{1}_A \otimes \mathbf{1}_B)(s) = \mu(A \cap (B + s)). \quad (1)$$

## Proof Intuition

The source already proves  $(i) \Rightarrow (ii) \Rightarrow (iii)$  using only the standing second-countable LCA hypotheses. In the reverse implication it also proves, without Euclidean structure, that weak-star closability forces

$$E_f \subseteq \text{null } P(\Gamma(U, V)).$$

The only remaining task is to find conull  $U_0 \subseteq U$  and  $V_0 \subseteq V$  such that no difference  $u - v$ , with  $(u, v) \in U_0 \times V_0$ , lies in this common zero set. Euclidean density balls do that in the source.

On a general second-countable LCA group, choose a nonnegative normalized  $L^1$  approximate identity. After a common subsequence, convolution with this approximate identity converges pointwise almost everywhere on every finite-measure piece in countable exhaustions of  $U$  and  $V$ . At a good pair  $(u, v)$ , both translated indicator averages eventually exceed  $1/2$ . Since the averaging kernel has mass one, the two translated sets must overlap in positive measure. Formula (1) then gives a tensor whose  $P$ -value at  $u - v$  is positive. This provides exactly the Cartesian-product, rather than merely product-almost-everywhere, nonvanishing needed by marginal nullity.

## Main Result

**Theorem 1.** *Let  $G$  be a second-countable locally compact abelian group, let  $f : G \rightarrow \mathbb{C}$  be measurable, put  $\varphi = Nf$ , and let  $U, V \subseteq G$  be measurable. The following are equivalent:*

1.  $(U \times V) \cap E_f^* \simeq \emptyset$ ;
2.  $\varphi|_{U \times V} \in \mathfrak{S}_{\text{loc}}(U, V)$ ;
3.  $\varphi|_{U \times V} \in \mathfrak{S}_{w^*}(U, V)$ .

Consequently Theorem 7.5 of the source paper holds throughout the standing second-countable LCA category, giving an affirmative answer to Problem 2.

## The Approximate-Identity Density Lemma

For measurable  $U, V \subseteq G$ , let  $\Gamma(U, V)$  denote the subspace of  $\Gamma(G)$  whose elementary tensors have first factor supported in  $U$  and second factor supported in  $V$ , and put

$$Z_{U,V} = \text{null } P(\Gamma(U, V)).$$

**Lemma 1.** *For arbitrary measurable  $U, V \subseteq G$ , there exist conull measurable subsets  $U_0 \subseteq U$  and  $V_0 \subseteq V$  such that*

$$(U_0 - V_0) \cap Z_{U,V} = \emptyset.$$

*More precisely, for every  $u \in U_0$  and  $v \in V_0$ , there are finite-measure sets  $U_k \subseteq U$  and  $V_l \subseteq V$  such that*

$$\mu(U_k \cap (V_l + u - v)) > 0.$$

*Proof.* Second countability and local compactness make  $G$  sigma-compact and its Haar measure sigma-finite. Choose increasing finite-measure exhaustions

$$U_1 \subseteq U_2 \subseteq \cdots \subseteq U, \quad V_1 \subseteq V_2 \subseteq \cdots \subseteq V$$

whose unions agree with  $U$  and  $V$  modulo null sets.

There is a sequence  $(a_n) \subseteq L^1(G)$  such that

$$a_n \geq 0, \quad \int_G a_n d\mu = 1, \quad T_n h := a_n * h \longrightarrow h \quad \text{in } L^1(G)$$

for every  $h \in L^1(G)$ . This standard sequential approximate identity can be seen directly: take normalized indicators of a countable shrinking base of relatively compact identity neighborhoods. Strong continuity of translations in  $L^1$  gives

$$\|a_n * h - h\|_1 \leq \int_G a_n(t) \|h(\cdot - t) - h\|_1 d\mu(t) \longrightarrow 0.$$

Enumerate the countable family

$$\mathcal{H} = \{\mathbf{1}_{U_k} : k \geq 1\} \cup \{\mathbf{1}_{V_l} : l \geq 1\} = \{h_1, h_2, \dots\}.$$

Choose a strictly increasing subsequence  $n_j$  such that

$$\sum_{m=1}^j \|T_{n_j} h_m - h_m\|_1 \leq 2^{-j}. \tag{2}$$

For fixed  $m$ , summing (2) over  $j \geq m$  and using Tonelli's theorem shows that

$$T_{n_j} h_m(x) \longrightarrow h_m(x) \quad \text{for almost every } x \in G.$$

Thus one and the same subsequence works almost everywhere for every member of  $\mathcal{H}$ .

Let  $U_k^0 \subseteq U_k$  and  $V_l^0 \subseteq V_l$  be the corresponding full-measure convergence sets, and set

$$U_0 = \bigcup_{k \geq 1} U_k^0, \quad V_0 = \bigcup_{l \geq 1} V_l^0.$$

These are conull in  $U$  and  $V$ . Fix  $u \in U_0$  and  $v \in V_0$ , and choose  $k, l$  with  $u \in U_k^0$ ,  $v \in V_l^0$ . Writing  $b_j = a_{n_j}$ , for all sufficiently large  $j$  we have

$$\int_G b_j(t) \mathbf{1}_{U_k}(u-t) d\mu(t) > \frac{1}{2}, \quad (3)$$

$$\int_G b_j(t) \mathbf{1}_{V_l}(v-t) d\mu(t) > \frac{1}{2}. \quad (4)$$

Put  $\alpha(t) = \mathbf{1}_{U_k}(u-t)$  and  $\beta(t) = \mathbf{1}_{V_l}(v-t)$ . Since  $\alpha + \beta \leq 1 + \alpha\beta$ , equations (3)–(4) and  $\int b_j = 1$  imply

$$0 < \int_G b_j(t) \alpha(t) \beta(t) d\mu(t).$$

Hence the set

$$D = \{t : u-t \in U_k, v-t \in V_l\}$$

has positive Haar measure. Translation invariance, with  $x = u-t$ , now gives

$$\mu(U_k \cap (V_l + u-v)) = \mu(D) > 0.$$

Because  $U_k$  and  $V_l$  have finite measure,  $\mathbf{1}_{U_k} \otimes \mathbf{1}_{V_l} \in \Gamma(U, V)$ ; by (1), its  $P$ -value at  $u-v$  is positive. Therefore  $u-v \notin Z_{U,V}$ , proving the lemma.  $\square$

## Proof of Theorem 1

*Proof.* The proof of Theorem 7.5 in [1] establishes  $(i) \Rightarrow (ii) \Rightarrow (iii)$  under the standing second-countable LCA hypotheses; no Euclidean density theorem is used in those implications.

Assume (iii). Before invoking density points, the source proof establishes

$$E_f \subseteq Z_{U,V} = \text{null } P(\Gamma(U, V)). \quad (5)$$

This step uses the density of the domain of the adjoint weak-star-closable multiplier and the source's identity  $fP(h) = P((Nf)h)$ , but no special structure of  $G$ .

Apply Lemma 1. It gives conull  $U_0 \subseteq U$ ,  $V_0 \subseteq V$  such that  $(U_0 - V_0) \cap Z_{U,V} = \emptyset$ . In view of (5),

$$(U_0 - V_0) \cap E_f = \emptyset.$$

Equivalently,  $(U \times V) \cap E_f^*$  is marginally null, which is (i). This completes the only implication left to prove.  $\square$

## Verification Notes

The proof was audited against pages 26–28 of the source PDF. The authors state in Remark 7.6(i) that the subgroup-of- $\mathbb{R}^n$ -or- $\mathbb{T}^n$  assumption is used only in  $(iii) \Rightarrow (i)$ , at the density-point step. The packet changes only that step.

The edge cases were checked as follows:

1. potentially infinite-measure  $U, V$  are handled by finite-measure exhaustions, ensuring the indicator tensors lie in  $L^2 \hat{\otimes} L^2$ ;
2. the summable-error choice (2) produces one common subsequence for all exhaustion pieces;
3. the conclusion holds on a Cartesian product of conull sets, the exact strength required for marginal nullity;
4. the sign and translation in (1) agree with the source convention  $P(\mathbf{1}_U \otimes \mathbf{1}_V)(s) = \mu(U \cap (V+s))$ ;
5. no metric, doubling, or differentiation-basis assumption remains.

No computational experiment is used or needed.

## Novelty Check, Scope, and Limitations

On August 9, 2026, bounded searches were run in arXiv and general web indexes for the exact source title, “Theorem 7.5,” “Problem 2,” “w\*-closable multiplier,” “local Schur multiplier,” and combinations with locally compact abelian groups and approximate identities. The searches found the source and the same authors’ later arXiv:1401.2620 on closable multipliers on group algebras. The later paper was inspected: its principal results concern multipliers on reduced group  $C^*$ -algebras and group von Neumann algebras, not the restricted  $U \times V$  equivalence here. No later solution of Problem 2 or this argument was located. The run’s cheap indexes likewise had no prior packet for arXiv:1001.4638. Novelty confidence is moderate pending a specialist citation search.

The result covers every group in Section 7’s standing category: second-countable LCA groups. The paper develops its multiplier theory over standard measure spaces, so non-second-countable LCA groups are outside the literal framework used there and are not claimed in this packet. The result answers only Problem 2; the other four source problems are unaffected.

## Human-Review Recommendation

Recommend expert verification as a candidate full solution. The principal review points are:

1. the common-subsequence construction in Lemma 1;
2. the positive-overlap inference from two averages exceeding  $1/2$ ;
3. the orientation of  $P$  and the finite-measure tensor placement;
4. the source-proof audit leading to (5);
5. the interpretation of Problem 2 within Section 7’s standing second-countability hypothesis.

## References

- [1] V. S. Shulman, I. G. Todorov, and L. Turowska, *Closable Multipliers*, Integral Equations and Operator Theory **69** (2011), 29–62, arXiv:1001.4638, doi:10.1007/s00020-010-1819-2.